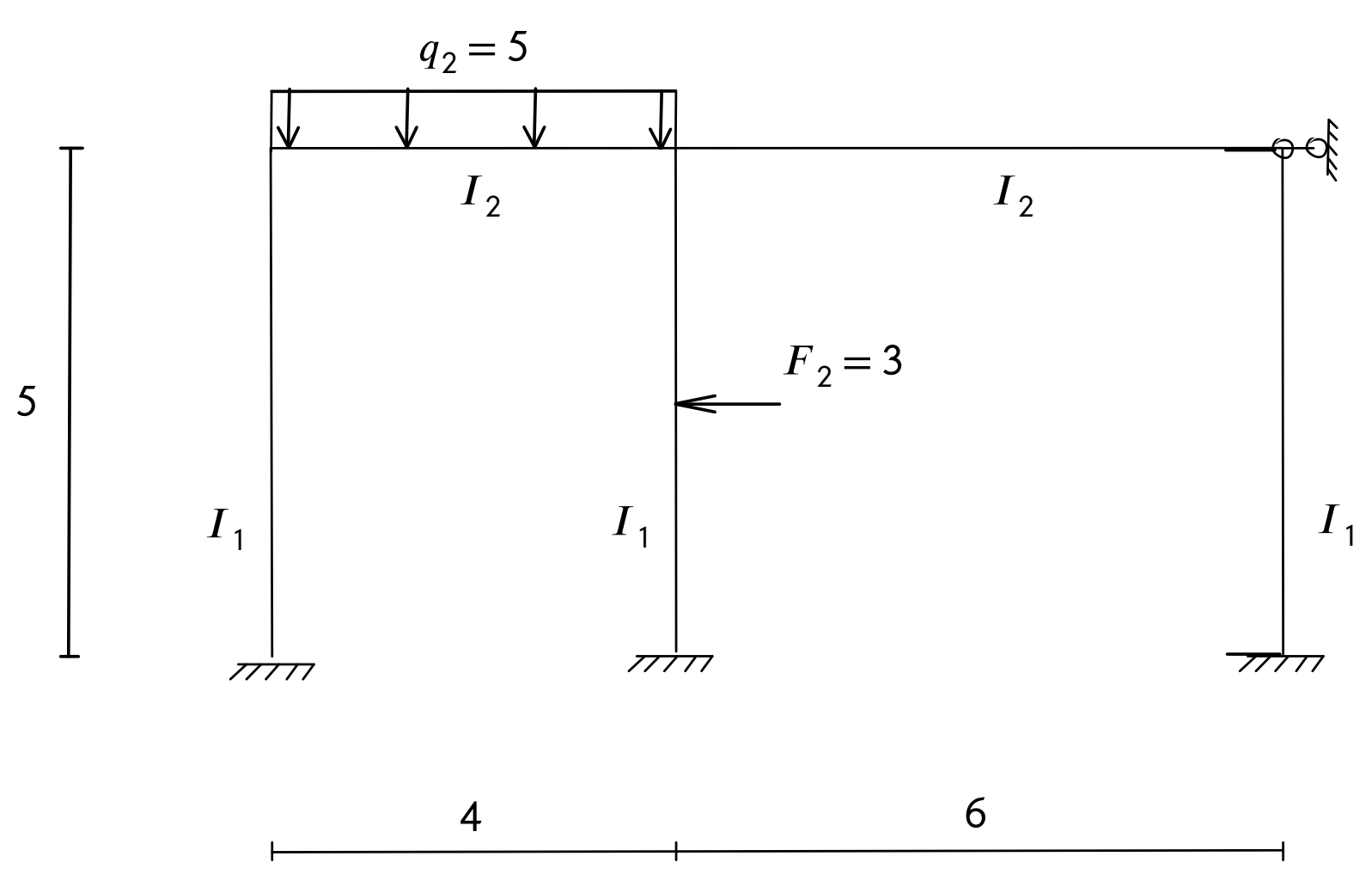
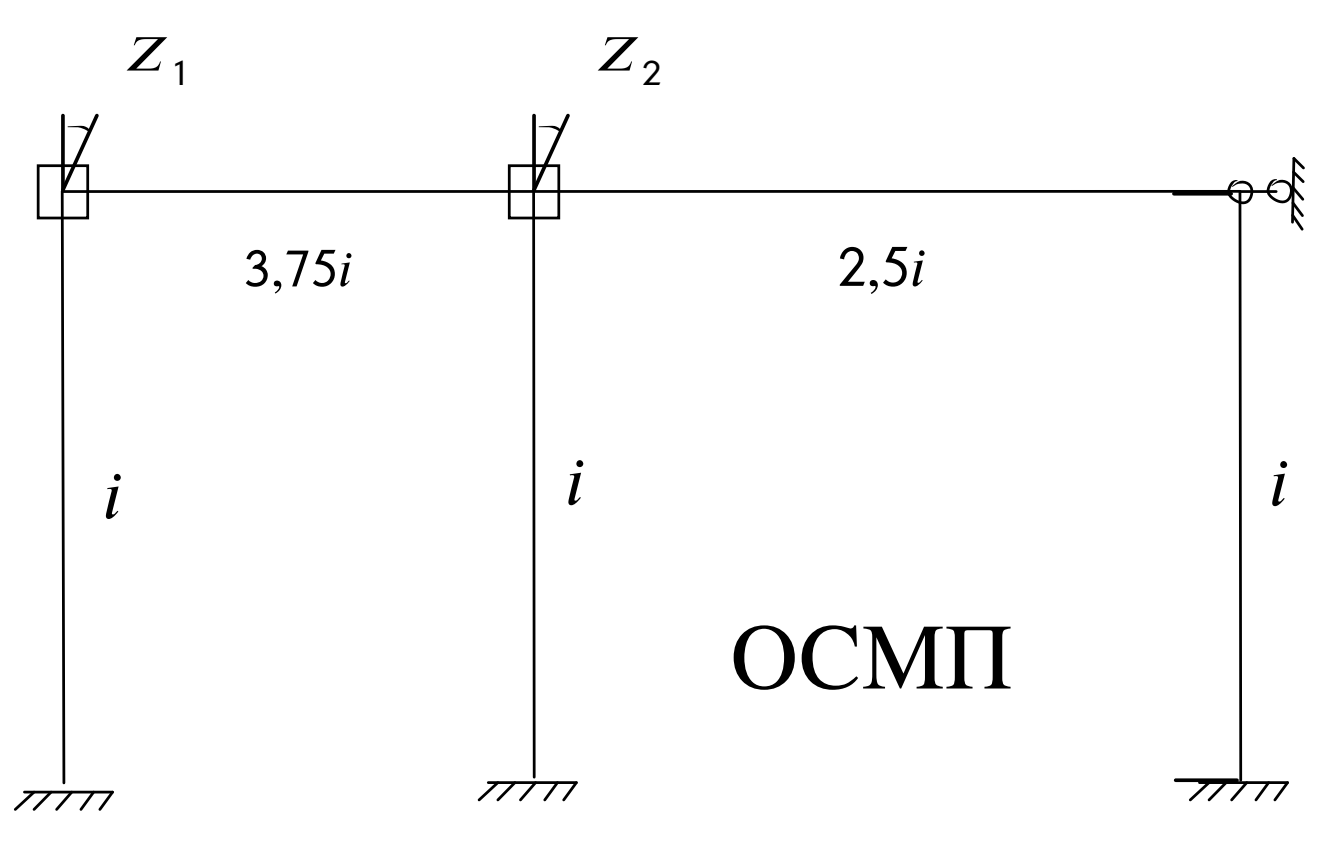


$\frac{I_2}{I_1} = 3$



ОСМП

Определим степень кинематической неопределимости

$$n_k = n_y + n_л = 2$$

Выбираем ОСМП

$\frac{I_2}{I_1} = 3 \gg EI_2 = 3EI_1$

$i = \frac{EI_1}{5}$

$i = \frac{EI_2}{l_1} = \frac{3EI_1}{4} = \frac{3*5}{4} = 3{,}75i$

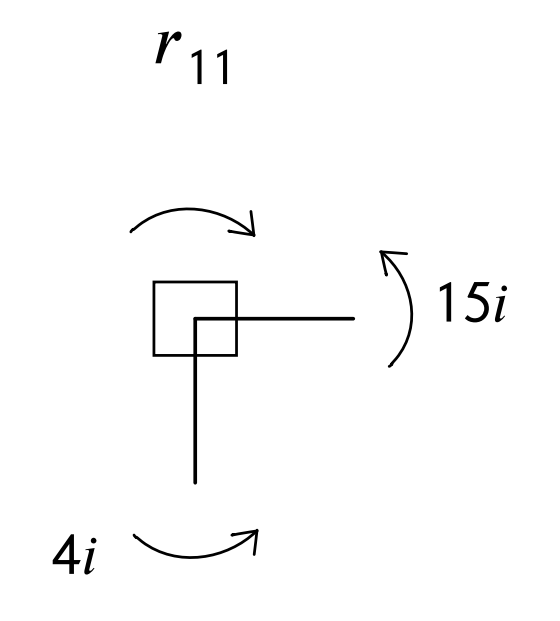
$i = \frac{EI_2}{l_2} = \frac{3EI_1}{6} = \frac{3*5}{6} = 2{,}5i$

Система канонических уравнений метода пермещения

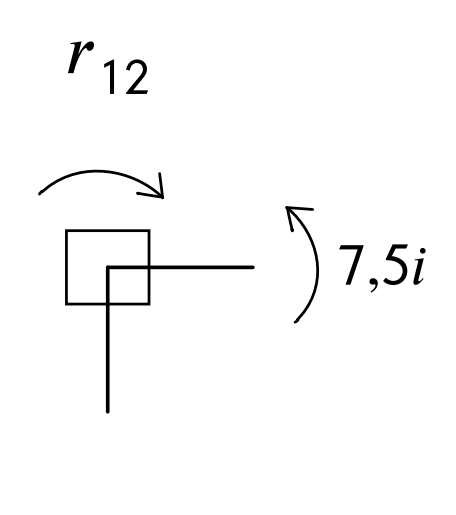
$$\begin{cases} r_{11}Z_1 + r_{12}Z_2 + r_{1F} = 0 \\ r_{12}Z_1 + r_{22}Z_2 + r_{2F} = 0 \end{cases}$$

Строим эпюры

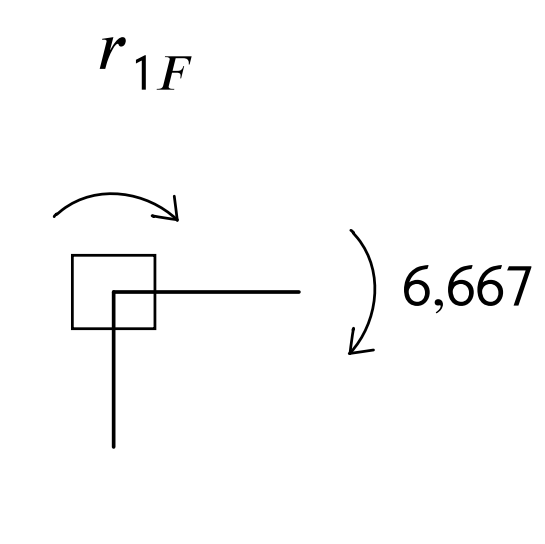
Определим реакции в доп связях



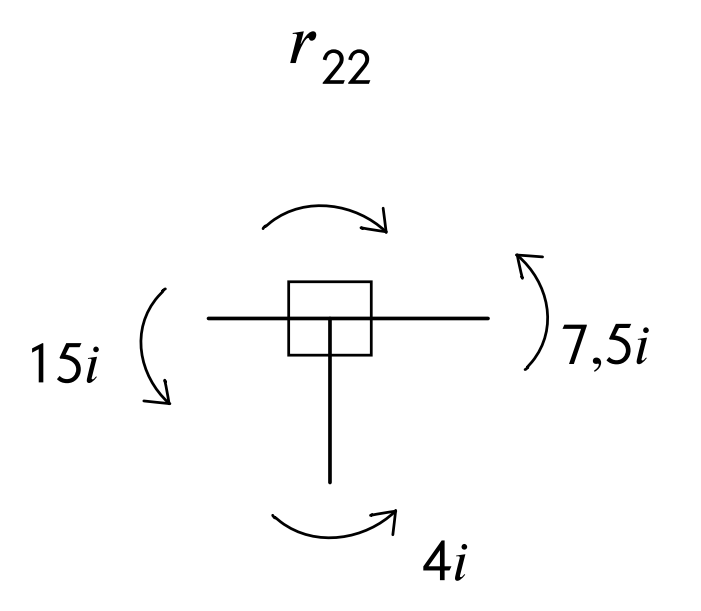
$r_{11} = 19i$



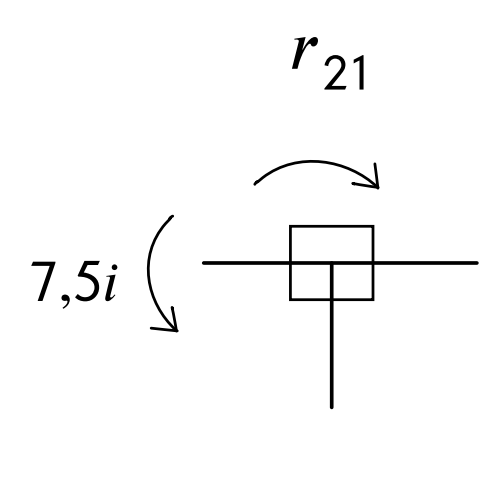
$r_{12} = 7{,}5i$



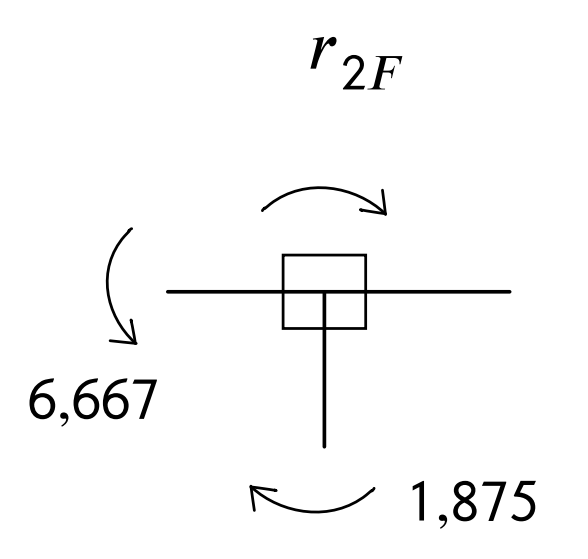
$r_{1F} = -6{,}667$



$r_{22} = 26{,}5i$



$r_{21} = 7{,}5i$

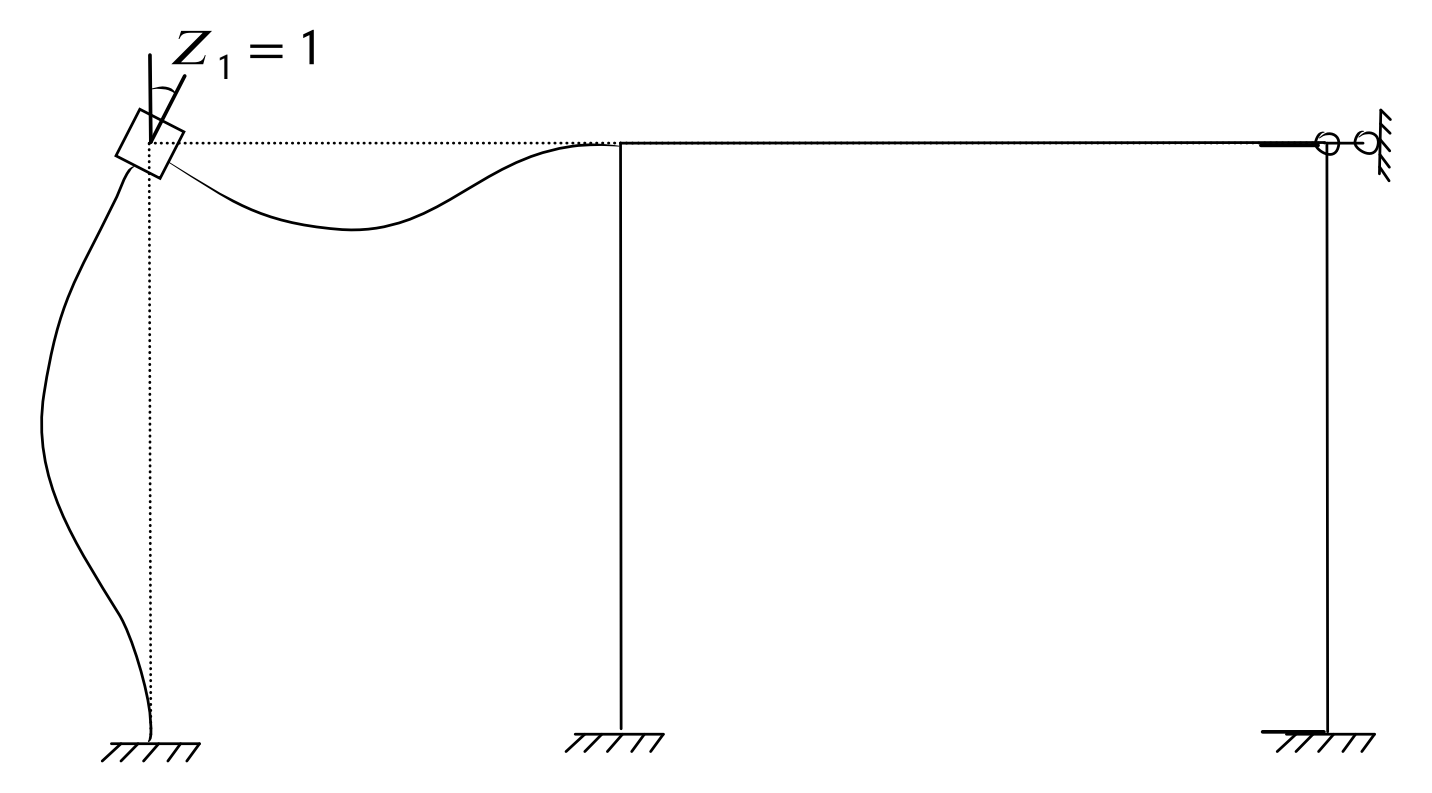


$r_{2F} = 4{,}791$

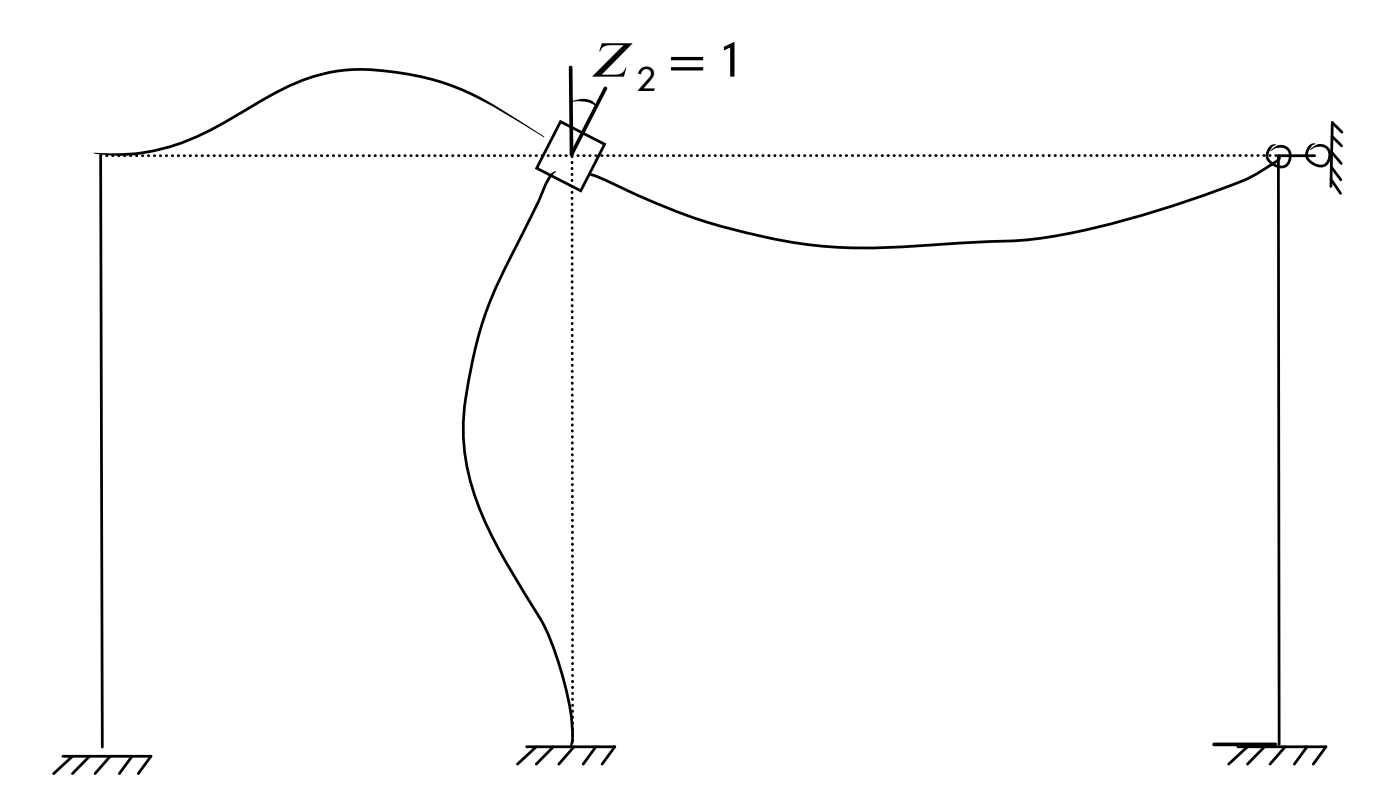
Решаем систему

$$\begin{cases} 19iZ_1 + 7{,}5iZ_2 - 6{,}667 = 0 \\ 7{,}5iZ_1 + 26{,}5iZ_2 + 4{,}791 = 0 \end{cases}$$

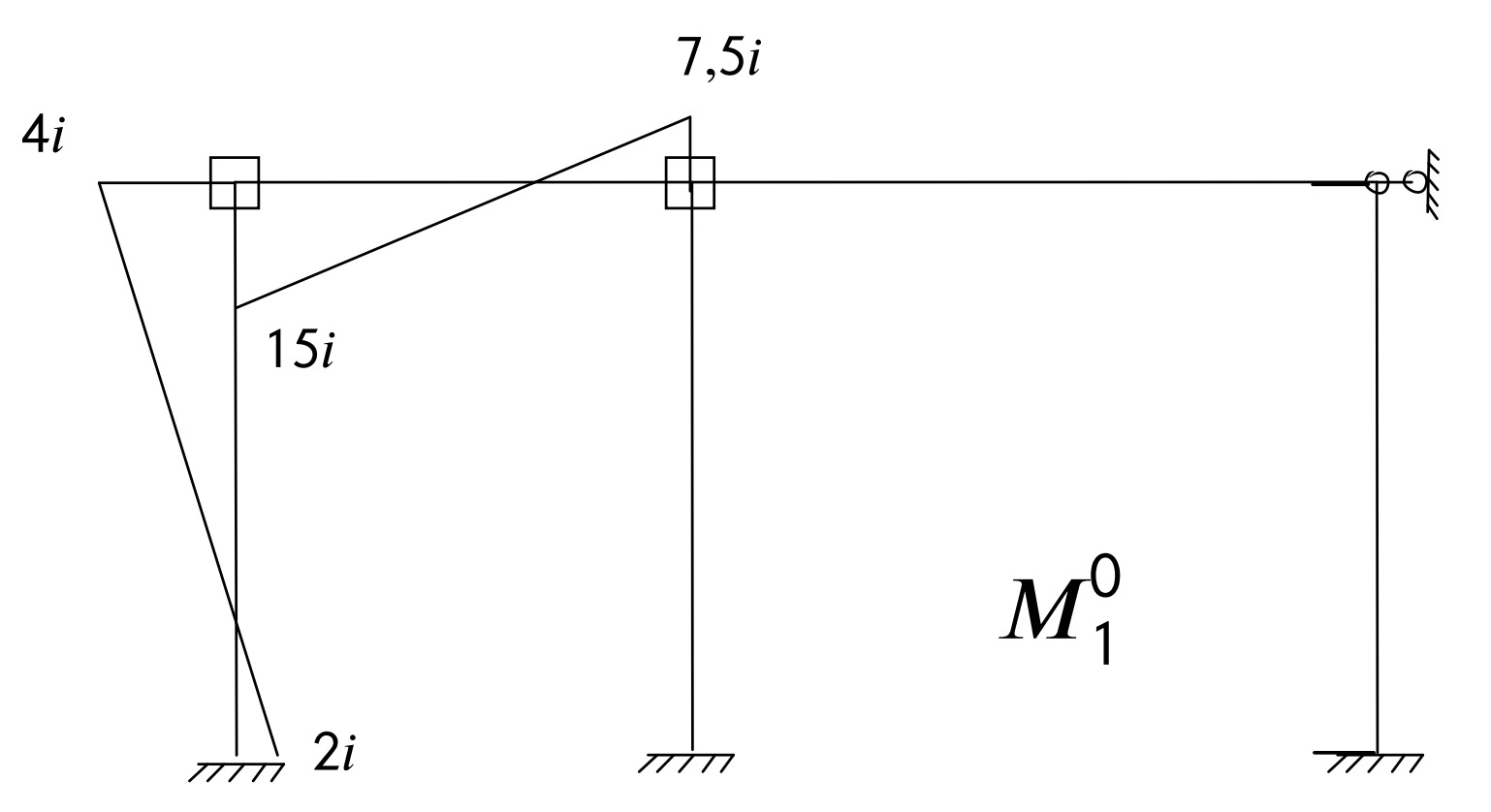
$$\begin{cases} Z_1 = \frac{0{,}4754}{i} \\ Z_2 = \frac{-0{,}3154}{i} \end{cases}$$

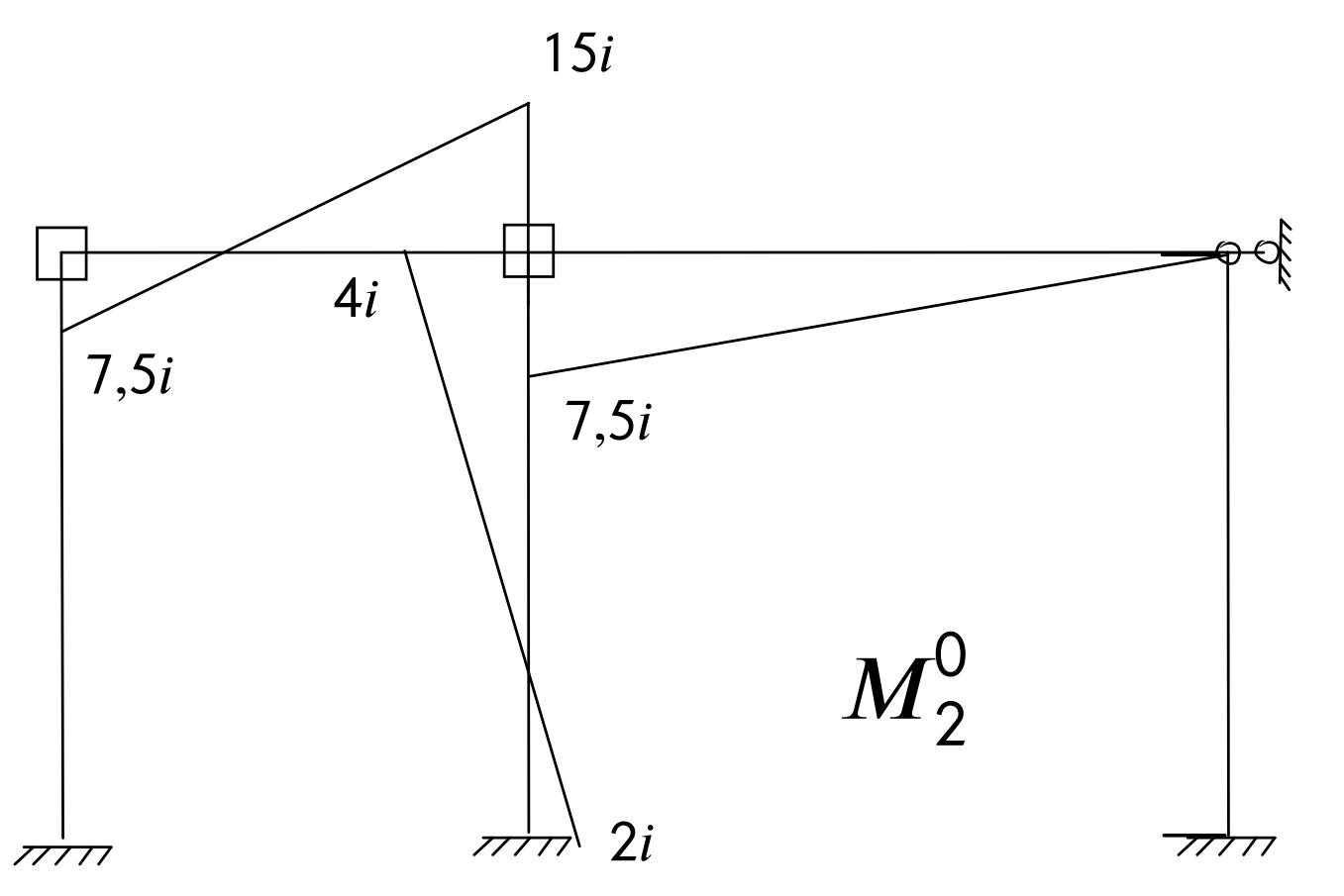


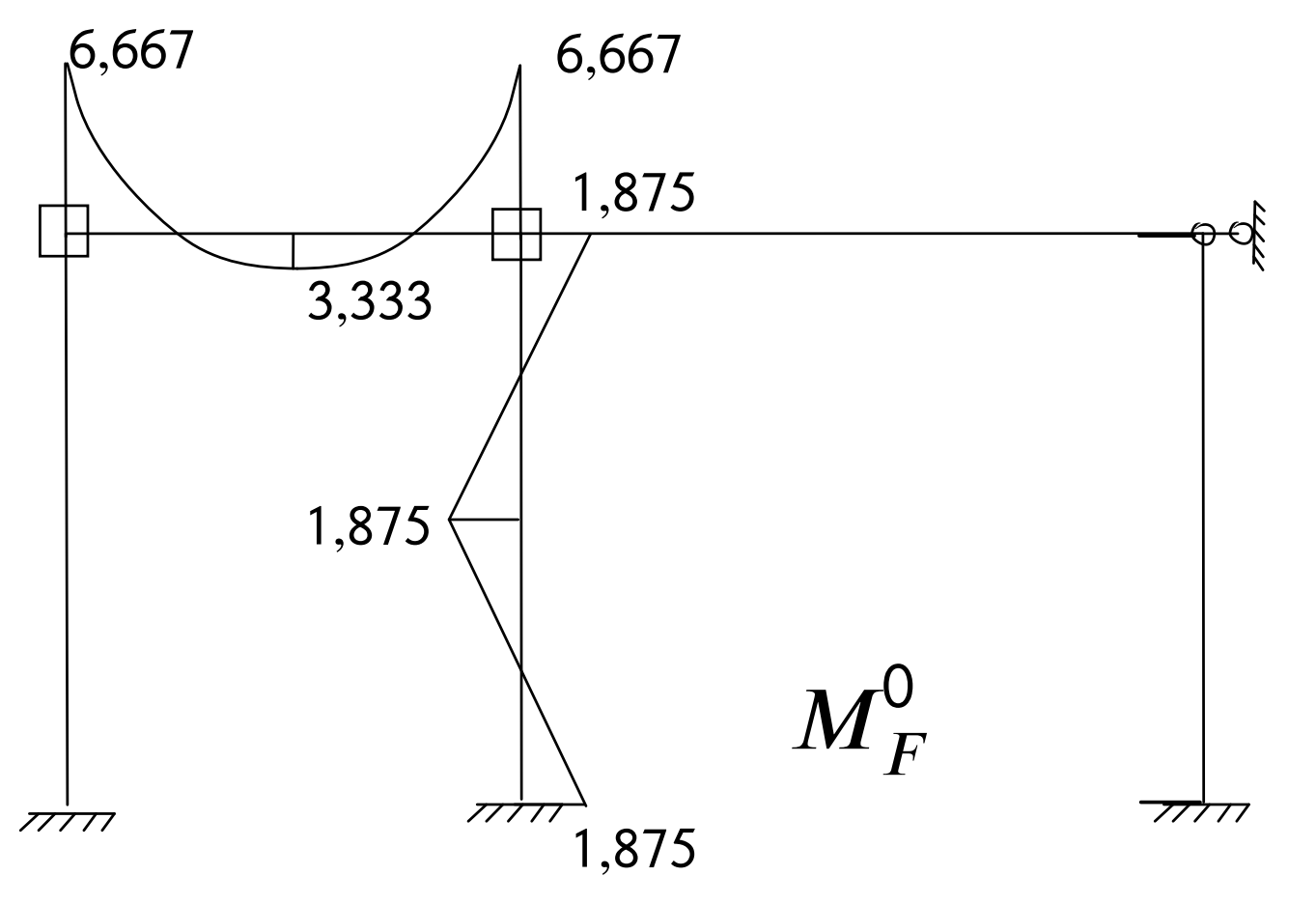
деф схема 1



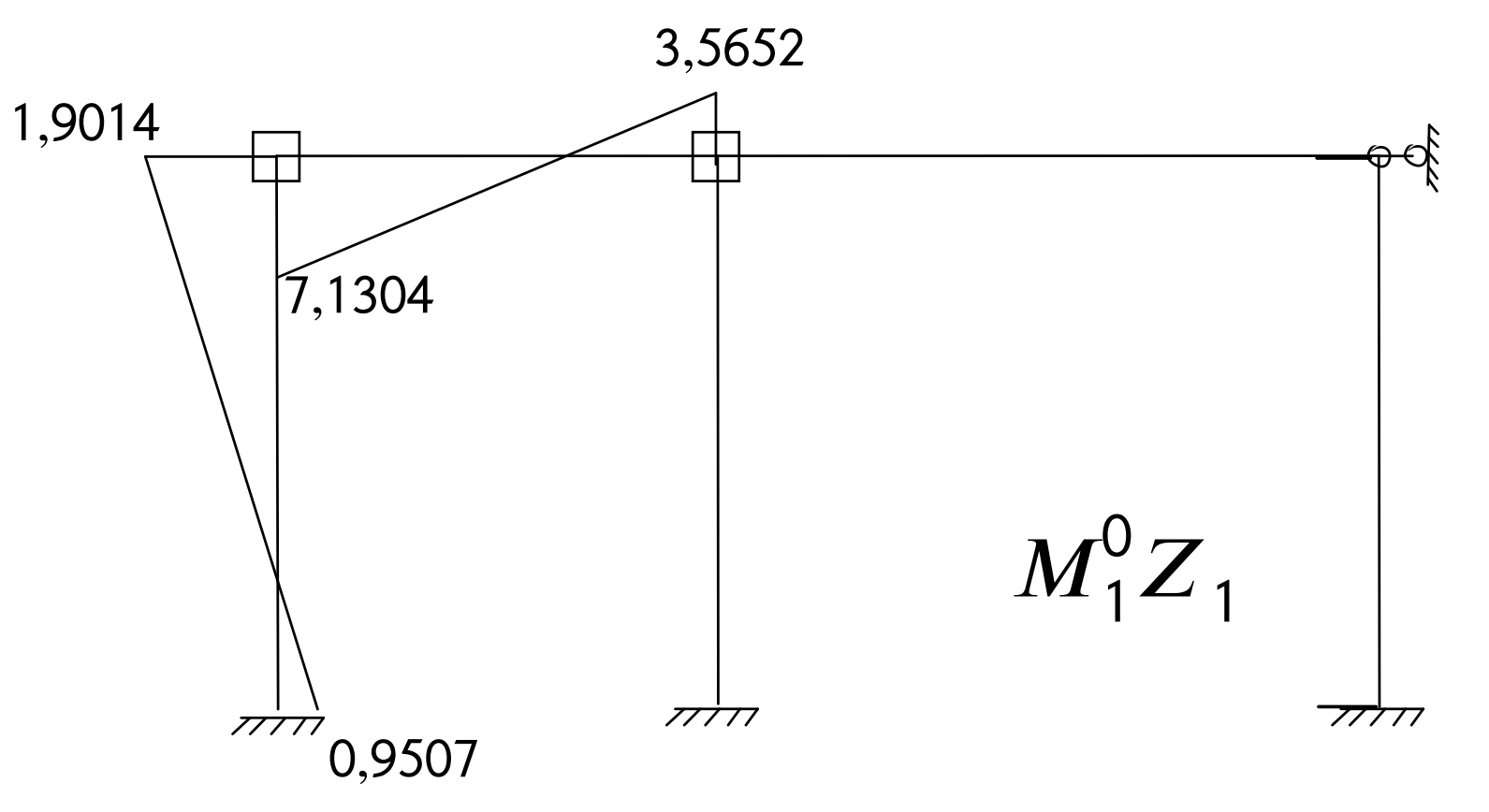
деф схема 2

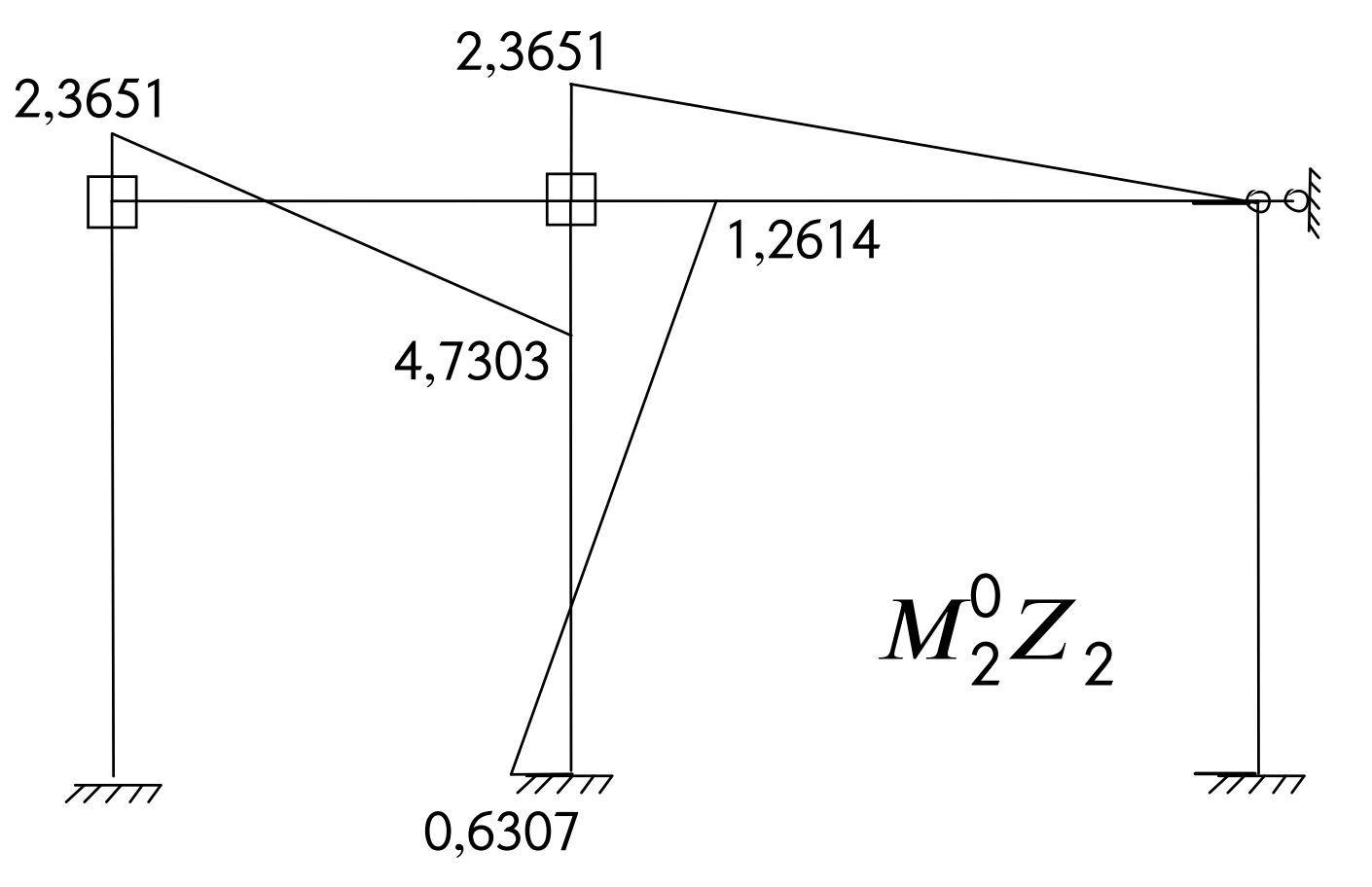




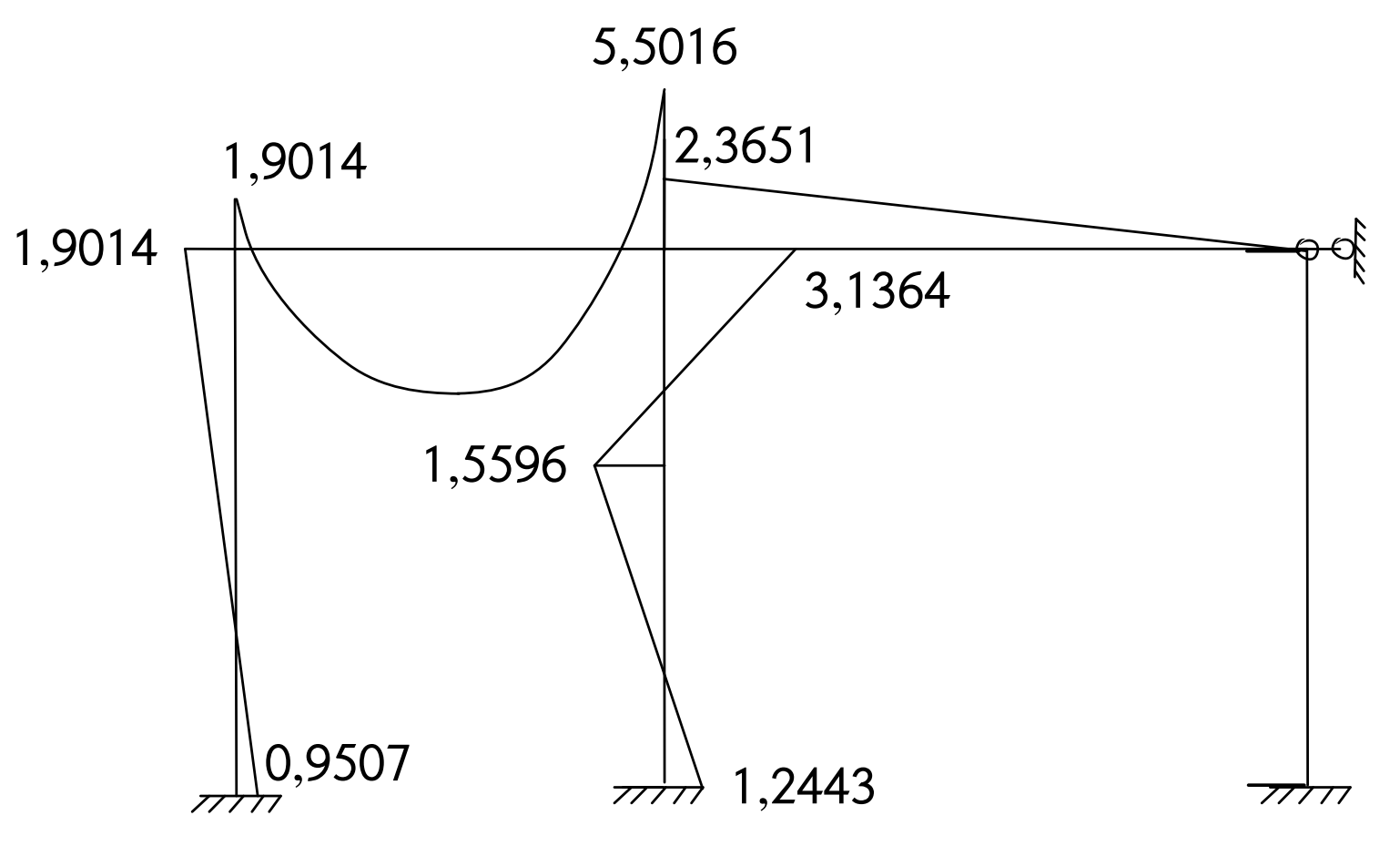


Построим исправленные эпюры



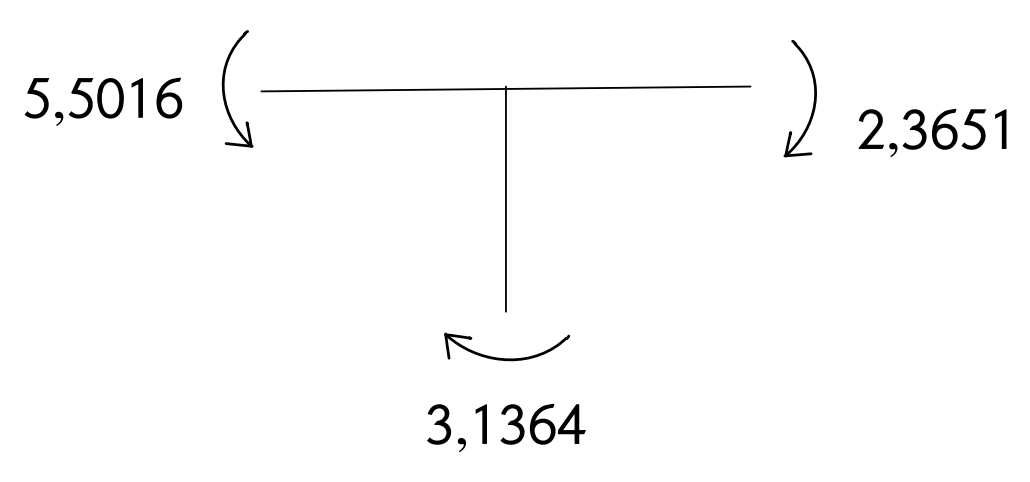


Построим конечную эпюру



M

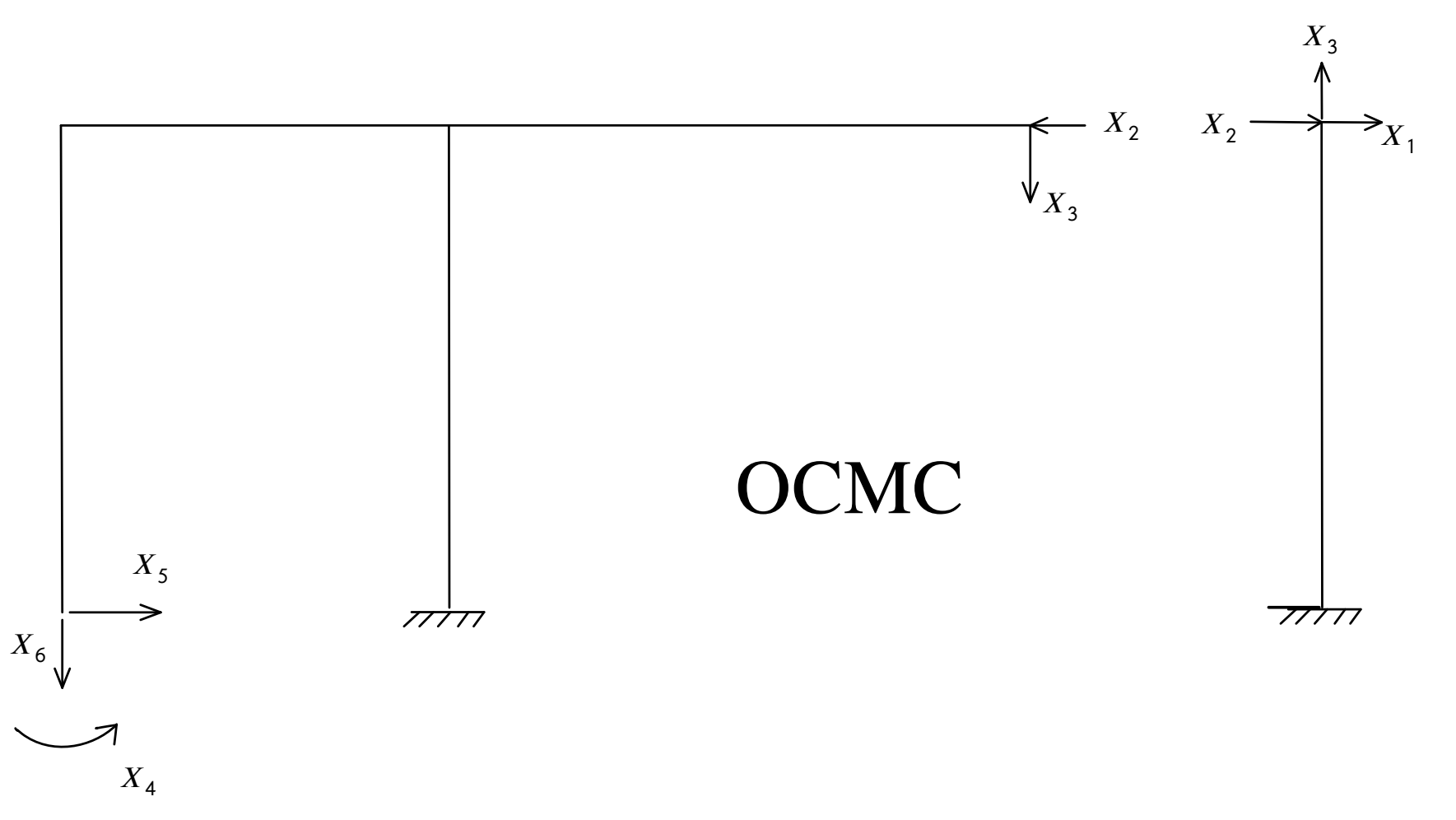
Равновесие узла



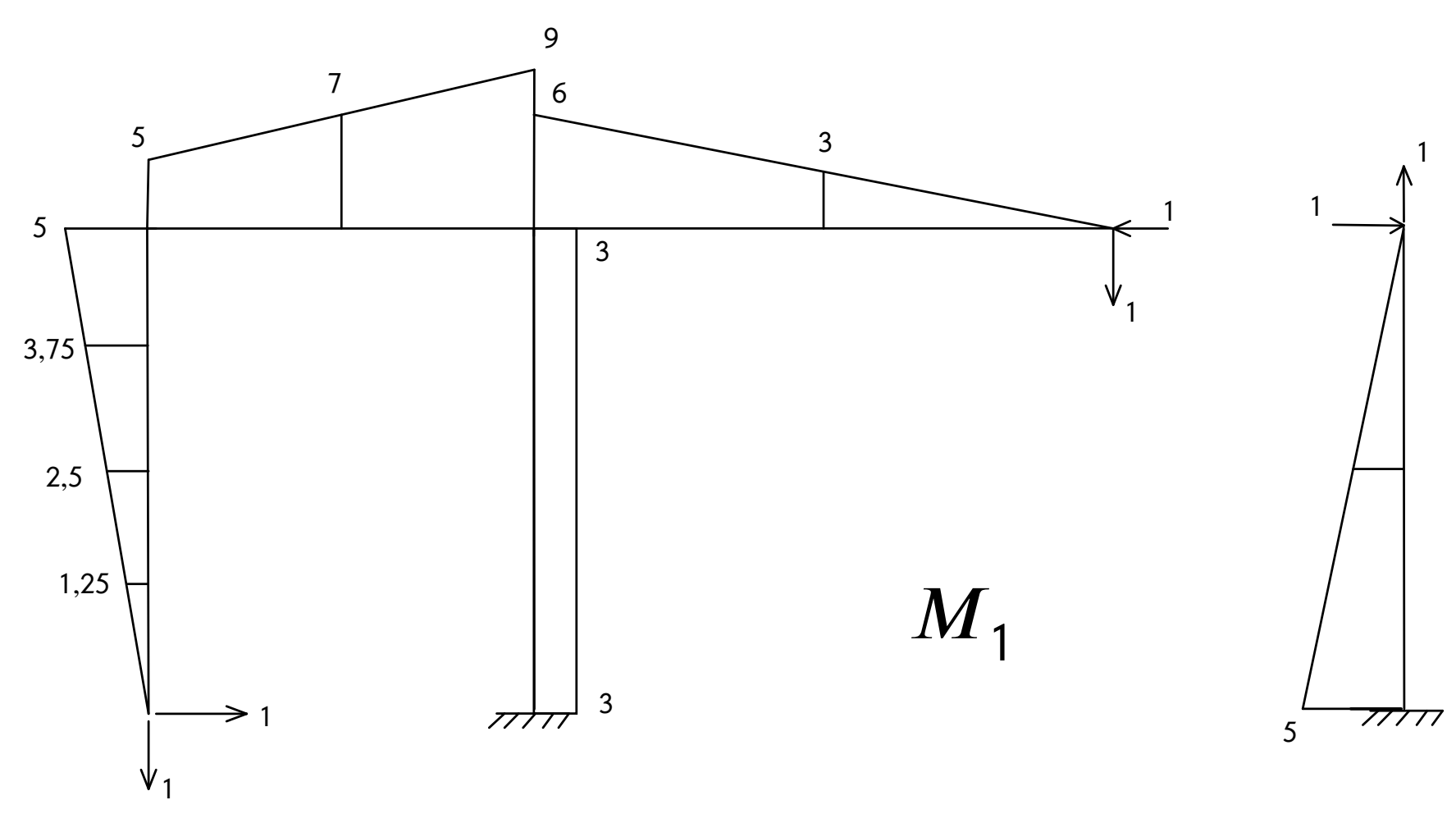
$0 = 0$

Деформационная проверка $n_{ст} = 3*3-3 = 6$

выбираем ОСМС

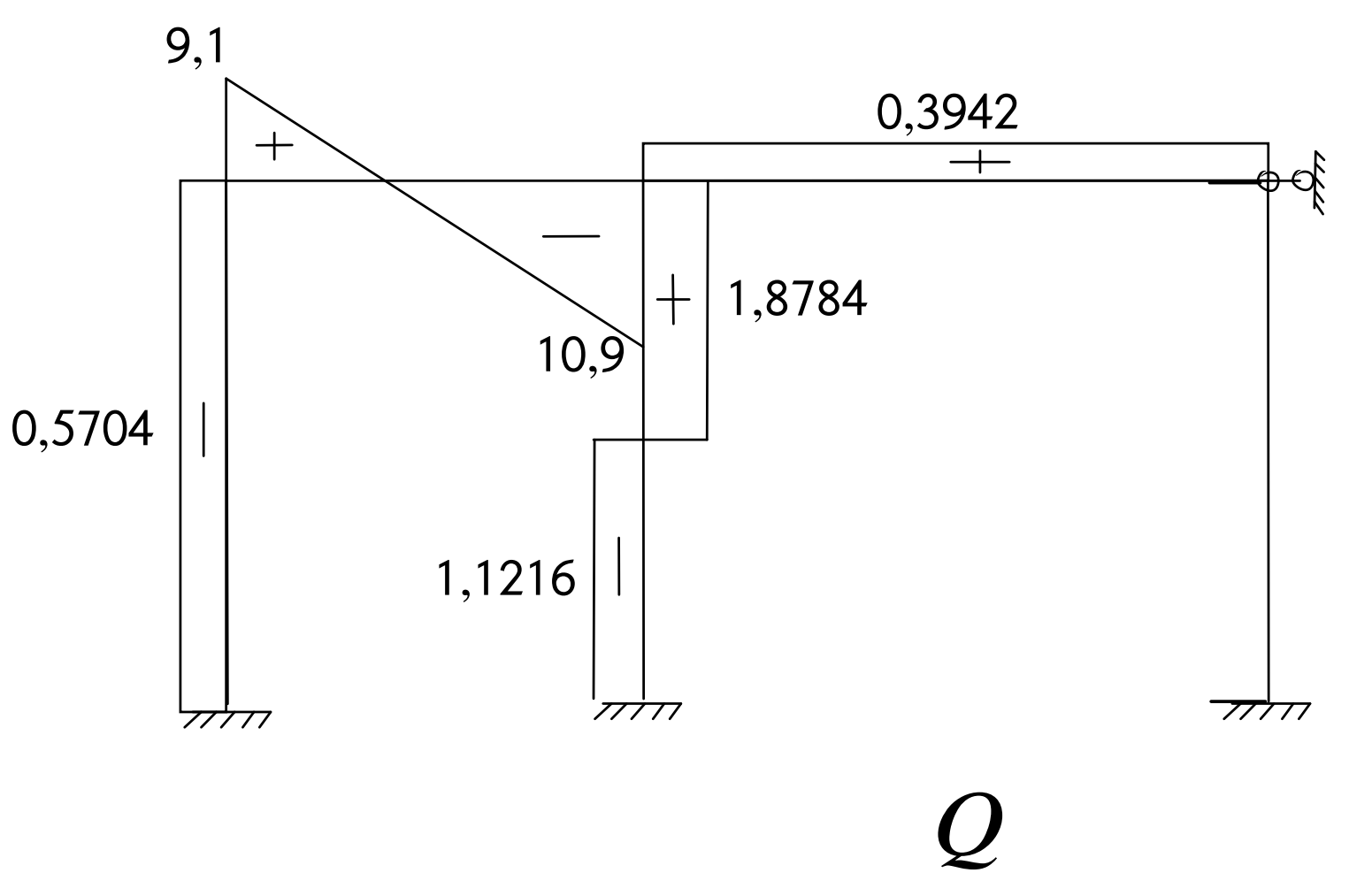


ОСМС

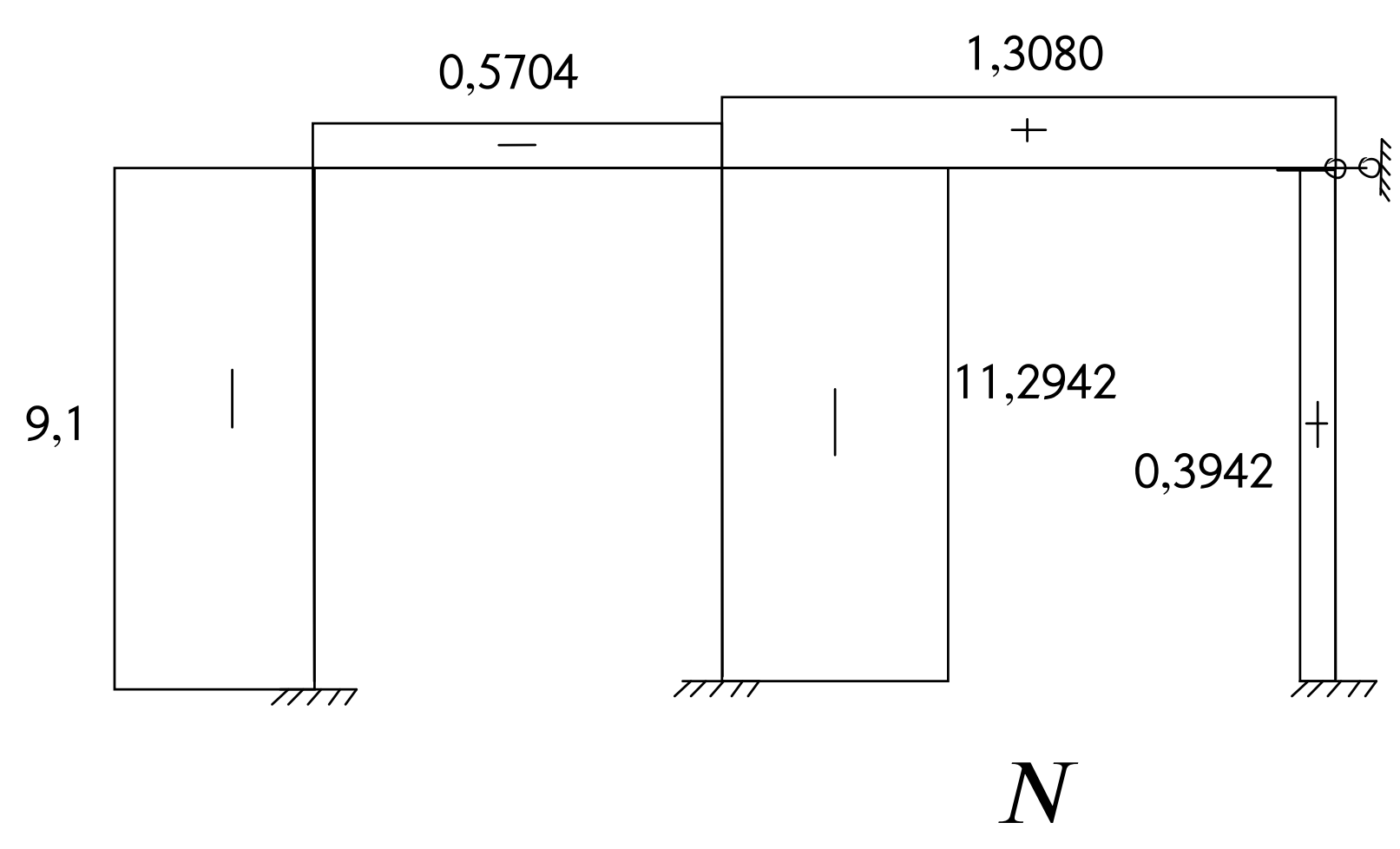


$$\Delta_{1F} = \frac{1}{EI}*\left(\frac{4}{6*3}(5*1{,}9014 - 4*6{,}2985*7 + 9*5{,}5016)\right) + \left(\frac{6}{6*3}(6*2{,}3651 + 4*3*1{,}1826)\right) + \left(\frac{2{,}5}{6}(3*3{,}1364 + 4*3*0{,}7884 - 3*1{,}5596)\right) + \left(\frac{2{,}5}{6}(-3*1{,}5596 - 3*4*0{,}1577 + 3*1{,}2443)\right) + \left(\frac{5}{6}(5*1{,}9014 + 4*2{,}5*0{,}4754)\right) = 0$$

Строим эп *Q N*

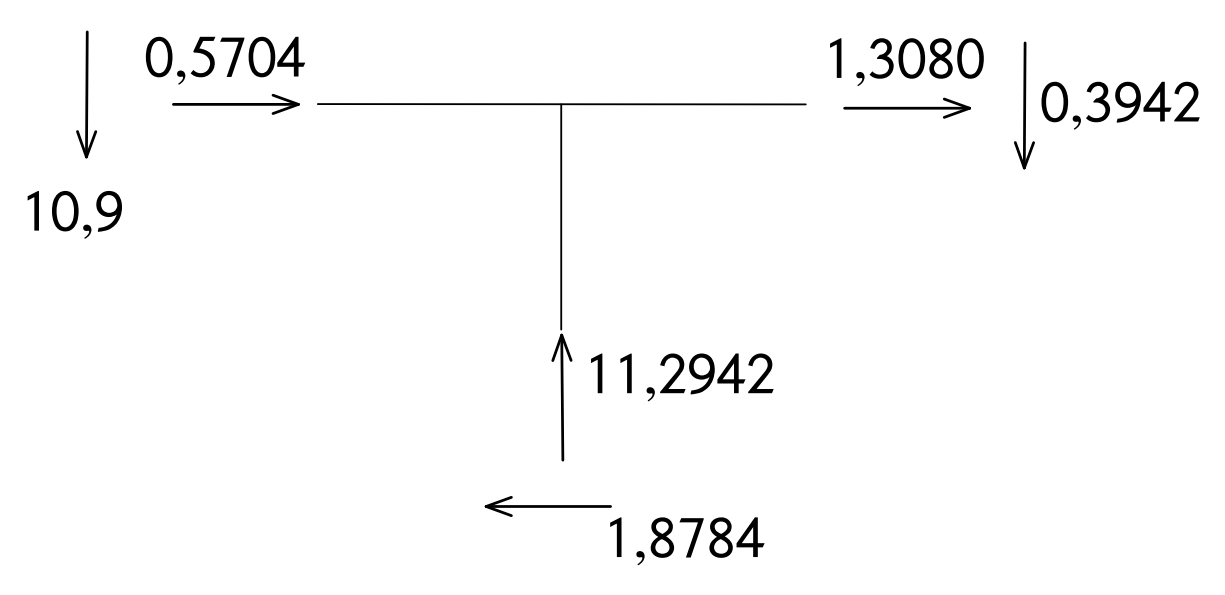


Q

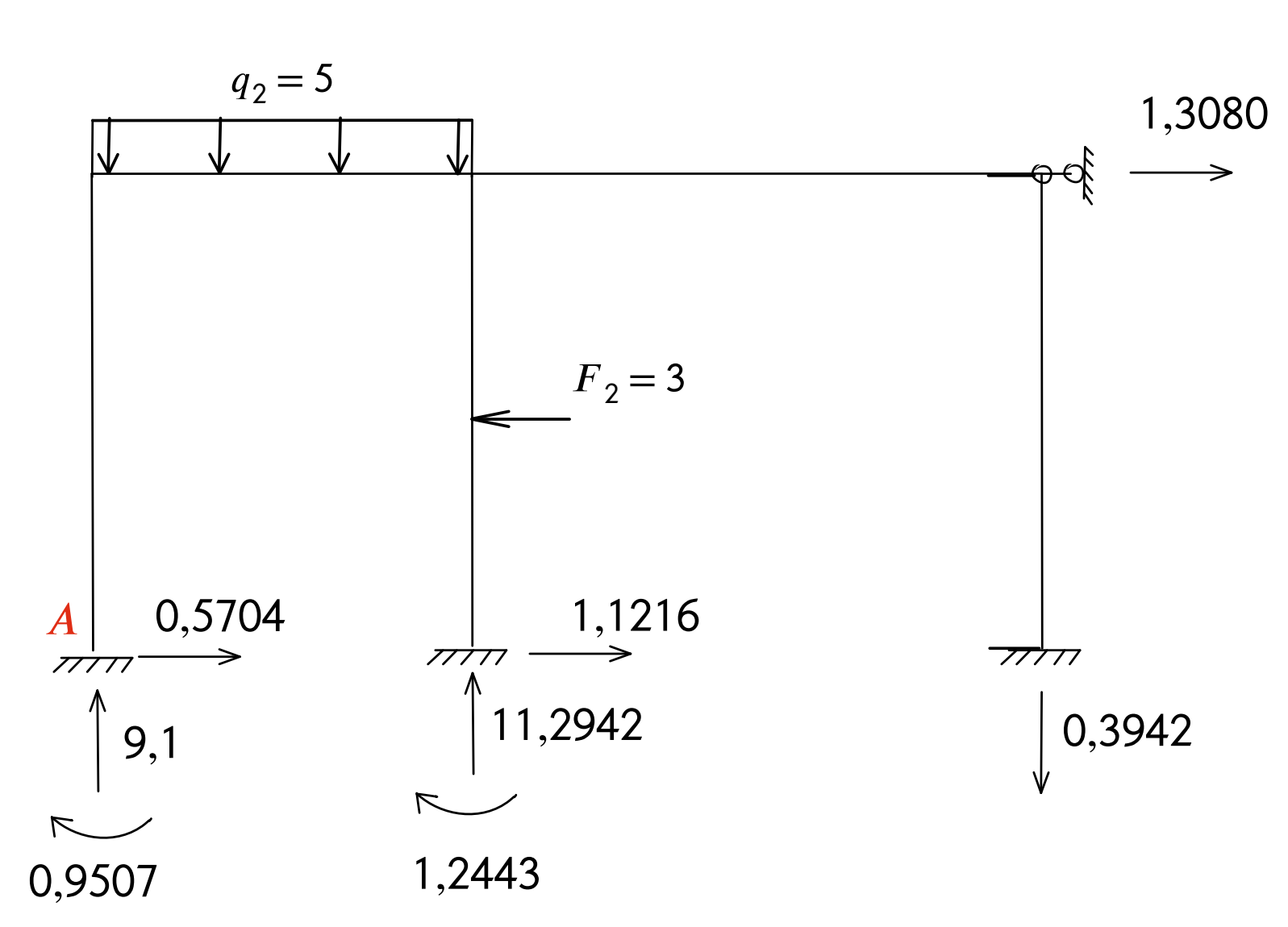


N

Равновесие узла



Определим реакции



$\sum X = 0 \qquad -3 + 1{,}3080 + 1{,}1216 + 0{,}5704 \approx 0$

$\sum Y = 0 \qquad 9{,}1 + 11{,}2942 - 0{,}3942 - 5*4 \approx 0$

$\sum M_A = 0 \qquad 0{,}9507 + 1{,}2443 + 5*4*2 - 3*2{,}5 - 11{,}2942*4 + 0{,}3942*10 + 1{,}3080*5 = 0$